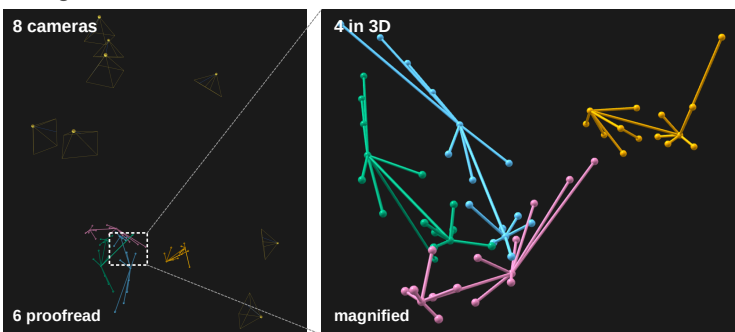
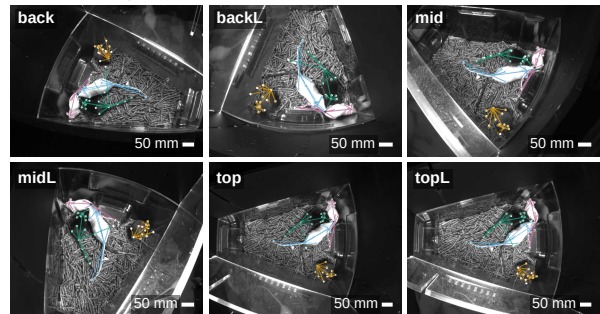


## a Rig and 3D

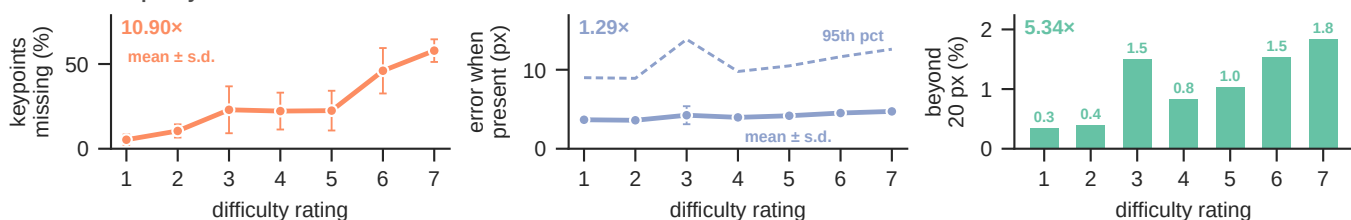


## b One frame, six cameras



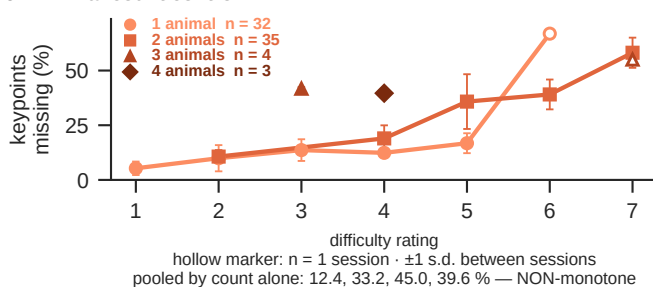
difficulty 4 · 4 animals · black bedding · bars 50 mm

## c Detection quality



$\pm 1$  s.d. between sessions · n = 12, 13, 9, 13, 10, 4, 13 sessions at difficulty 1–7

## d Animal-count control



## e Corpora

|                     | BMimica       | SLAP-2M         | Total      |
|---------------------|---------------|-----------------|------------|
| Cameras             | 5             | 8 (6 proofread) | —          |
| Animals             | 2             | 1–4             | 1–4        |
| Sessions with 3D    | 56 of 56      | 74 of 84        | 130 of 140 |
| Frames              | 10,084,734    | 1,954,440       | 12,039,174 |
| Hours               | 18.7 @150 fps | 10.9 @50 fps    | 29.5       |
| Nodes               | 15            | 15              | 15         |
| Measured in c, d, f | no            | yes             | no         |

## f Difficulty strata

| Difficulty | Sessions | Keypoints | Animals | Bedding (black/white) | Missing (%) | Error mean (px) | Error p95 (px) | Error p99 (px) | > 20 px (%) |
|------------|----------|-----------|---------|-----------------------|-------------|-----------------|----------------|----------------|-------------|
| 1          | 12       | 252,265   | 1       | 12/0                  | 5.3         | 3.67            | 9.00           | 14.52          | 0.34        |
| 2          | 13       | 342,287   | 1, 2    | 8/5                   | 10.5        | 3.60            | 8.91           | 14.67          | 0.39        |
| 3          | 9        | 229,072   | 1, 3    | 4/5                   | 23.0        | 4.24            | 13.85          | 21.58          | 1.50        |
| 4          | 13       | 245,731   | 1, 2, 4 | 6/7                   | 22.2        | 3.97            | 9.78           | 19.04          | 0.83        |
| 5          | 10       | 273,748   | 1, 2    | 6/4                   | 22.5        | 4.17            | 10.49          | 20.88          | 1.03        |
| 6          | 4        | 91,922    | 1, 2    | 1/3                   | 46.0        | 4.52            | 11.65          | 24.61          | 1.53        |
| 7          | 13       | 126,890   | 2, 3    | 7/6                   | 57.9        | 4.72            | 12.60          | 27.36          | 1.83        |

a, b: one SLAP-2M session. c, d, f: 74 SLAP-2M sessions, 1,561,915 keypoints, raw detections from the benchmark's shared identity-stripped pool, matched per frame against the proofread 3D reprojected into each camera. e: both corpora, read from the files.